FLOWCHART TEKNIS — SMARTESC STM32 V2

rofiqcp/SmartESC_STM32_v2 • branch vesc_comp • m365/Src/main.c • STM32 HAL + FreeRTOS

I. INISIALISASI

- START
- main() → HAL_Init(); SystemClock_Config()
- Init GPIO / DMA / ADC / TIM / UART
- MX_NVIC_Init()
- ADC1/ADC2 calibration; MX_MotorControl_Init()
- conf_general_init(); task_init()
- Start scheduler

II. MAIN LOOP / TASK

- **xTaskCreate(startMediumFrequencyTask, PRIO_NORMAL)**
- xTaskCreate(StartSafetyTask, PRIO_HIGHER)
- vTaskStartScheduler()
- Task control and safety services
- Motor setup / command via VESC interface
- Repeat tasks while scheduler active

III. ISR / KONTROL

- **TIM1_UP_IRQn priority (0,0)**
- ADC1_2_IRQn priority (2,0)
- TIM3_IRQn priority (3,0)
- TIM1_BRK_IRQn priority (4,0)
- DMA1_Channel1_IRQn priority (0,0)
- ADC injected channels 3/4/5 triggered by TIM1_CC4

IV. FAULT & SAFETY

- **Init error? → Error_Handler()** (TIDAK / YA)
- **Fault code / safety task?** (TIDAK / YA)
- MC_FOC_DURATION / MC_OVER_VOLT
- MC_UNDER_VOLT / MC_OVER_TEMP
- MC_START_UP / MC_SPEED_FDBK
- MC_BREAK_IN (overcurrent) / MC_SW_ERROR

DETAIL SOURCE CODE / CATATAN

Identifier & register terverifikasi di m365/Src/main.c: main(), SystemClock_Config(), MX_GPIO_Init(), MX_DMA_Init(), MX_ADC1_Init(), MX_ADC2_Init(), MX_TIM1_Init(), MX_TIM3_Init(), MX_USART3_UART_Init(), MX_USART1_UART_Init(), MX_NVIC_Init(), MX_MotorControl_Init(), conf_general_init(), startMediumFrequencyTask(), StartSafetyTask(), task_init(), Error_Handler(). Nama ISR kontrol FOC/rumus Clarke-Park harus ditelusuri di modul motor-control, tidak disimpulkan dari main.c.